

From: Ed May ed@bldgtyp.com
Subject: PHIUS Non-Res Loads, Schedules ?
Date: September 24, 2021 at 11:33 AM
To: Al Mitchell amitchell@phi.us.org



Hi Al,

I hope that you are well. Great to talk with you a bit back - and glad to hear that you are interested in the stuff that Trevor and I have been working on. If you have a minute, I wonder if I might ask your opinion on something regarding Non-Residential modeling and see if you have any thoughts on it?

Some background:

1. As you know, modeling Non-Res occupancy, lighting and equipment usage is challenging. One of the things that I would like to do with our new Rhino tooling is make that easier, and also to open up the really nice library of existing space-types (loads and schedules) to WUFI users in the future.
2. In the Ladybug tool, currently they have already built in many, many [space types](#) into their software, all drawn from OpenStudio, as far as I know (so: DOE data). However, the format of this data is significantly different from the current WUFI configuration, and so one thing I'm working on is finding a translation mechanism so that we can use these libraries.
3. These Ladybug/OpenStudio libraries all break down the 'Program' of the space into three basic sectors:
- Occupancy
 - Lighting
 - Elec. Equipment Usage

each of these sectors then has:

- Load(s)
- Schedule

This method makes things quite a bit easier to manage, and allows for a lot more customization / organization, I think. It also makes the modeling more predictable, which is nice.

4. So, for translating all that into 'WUFI-speak', my thinking is that we can simply build out Utilization Patterns for each of the above Schedules (Occupancy, Lighting, Elec. Equip) for each space-type used. So for instance, we can get something like:

WUFI® Passive V.3.2.0.1

\\Mac\Home\Dropbox\bldgtyp-00\00_PH_Tools\PyPH\sample\EM_sample_output_WUFI_XML\Sample_Output_9_23_14_42_39.xml

FileInputOptionsDatabaseHelp

ScopePassive house verificationEnglish/SI/Outer dimensions/PHIUS+ 2018Assign data

Internal Loads/Occupancy

Ventilation/Rooms

Attached zones

Remaining elements

Systems

System 1 (User defined)

Device 1 (Mechanical ventilation: Ventilation): A F

Case 2: Non-Residential

Localization/Climate: User defined

Building

PH case: Passive house: Office/Administrative buildir

Zone 1: Merged Zone

Visualized components

Component 1: INT_207_TLT

Component 2: EXT_207_TLT

Component 3: INT_207_TLT

Component 4: EXT_207_TLT

Component 5: EXT_208_Pantry

Component 6: Aperture_7f9d9f99

Component 7: EXT_107_TLT

Not visualized components

Thermal bridges

Internal Loads/Occupancy

Ventilation/Rooms

Attached zones

Remaining elements

Systems

System 1 (User defined)

Device 1 (Mechanical ventilation: Ventilation):

Utilization pattern

OccupancyOffice equipmentKitchen equipmentLightingProcess loads

Name	Begin utilization [h]	End utilization [h]	Annual utilization days [days/a]	Illumination level [lux]	Height of utilization level	Relative absence [-]	Part use factor of operating period for lighting [-]
_default_space_occupa	0	24	365	300		0.31	1.0
OCC_PublicRestroom	0	24	365	0	Level 1: 0 m	0.73	0.66
LGHT_PublicRestroom	0	24	365	0	Level 1: 0 m	56	.98
OCC_Stair	0	24	365	0	Level 1: 0 m	0.00	0.55
LGHT_Stair	0	24	365	300	Level 1: 0 m	0.40	.32
OCC_Kitchen	0	24	365	0	Level 1: 0 m	0.42	0.34
LGHT_Kitchen	0	24	365	0	Level 1: 0 m	0.34	.85

Optional data: LGHT_PublicRestroom

Average occupancy [m²/Person]	
Room setpoint temperature [°C]	
Heating reduction temperature [K]	
Daily utilization hours [hrs/d]	
Annual utilization hours [hrs/a]	
Annual utilization hours during daytime [hrs/a]	
Annual utilization hours during nighttime [hrs/a]	
Daily heating operation hours [hrs/d]	
Daily ventilation operation hours [hrs/d]	

NewDeleteCopyInsertNew/Insert:after

These are loads, not schedules? Odd that these are here.

Where we have separate Occupancy, Lighting, and Elec. Equipment Schedules (Util. Patterns) for each space type in the model, which we then use in the later WUFI tabs.

Hopefully all the above makes sense? So, I wonder what you think about:

1. Doing this will result in more Utilization Patterns. We'd have three for each space type (well, 4 if you include the Ventilation Schedule for the space type as well...). Which is nice from a customization / control / accuracy perspective. But I wonder if you see a downside from this? Obviously more patterns means more to check and verify. But I actually think splitting apart the schedules in this manner probably makes it easier (I hope?) to check / adjust the various bits, as opposed to now where the schedules are all sort of joined together and there isn't clear separation between them?
2. The OpenStudio schedules and loads are all hourly. So they don't map to the 'start hour', 'end hour' WUFI thing very well. So my thought is that we can just set the start / end to 0 and 24 respectively, and 365 days, then calc the annual util factor based on the OpenStudio hourly values, and input that as the use-factor (or absence factor).

Do you think you see any significant issues with entering data this way? The start/end hours aren't important, other than for helping to define the use-pattern, are they? Do you think the reviewers will be unhappy if we don't include those somehow? Personally, I think just entering an annual factor makes a lot more sense than entering a factor of the use-period, as defined by start/end? But if you think those values are super important, then I can rethink that?

In general though, we would then have an easy method for taking 8670 hourly use-values, and turning those into an average annual-use-factor. This means we can then just use all those OpenStudio libraries which is nice. So long as we don't *have* to use the 'start hour/end hour' method of defining the use-period?



